Lecture 4 Summary: Computer Architecture & IAS Machine

This lecture focuses on the **difference between Computer Architecture and Computer Organization**, and a detailed study of the **IAS Machine (Von Neumann Architecture)**.
Dr. Shadenda frequently asks **very precise MCQ and True/False questions**, especially about **IAS registers**.

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1. Computer Architecture vs. Computer Organization

Computer Architecture

- Refers to attributes **visible to the programmer**.
- Describes **WHAT the computer does**.
- Has a **direct impact on program execution**.
- Examples:
    - Instruction Set
    - Data representation
    - Number of bits used for data

Computer Organization

- Refers to hardware details **transparent to the programmer**.
- Describes **HOW the computer works internally**.
- Concerned with operational units and their interconnections.
- Examples:
    - Control signals
    - Memory technology
    - Internal hardware implementation

📌 **Exam Tip**

- Architecture = **WHAT**
- Organization = **HOW**

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2. IAS Machine (Von Neumann Architecture)

The IAS machine is the **core model** used in this lecture.

Memory

- Memory size: **1024 words**
- Word size: **40 bits**

Word Format

Each 40-bit word can store:

- **One 40-bit data value**, OR
- **Two instructions**

Instruction Layout:

- Left Instruction: bits **0–19**
- Right Instruction: bits **20–39**
- Each instruction consists of:
    - **8-bit Opcode**
    - **12-bit Address**

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3. IAS Registers (Very Important for Exam)

You must memorize **each register name and function**.

MBR (Memory Buffer Register)

- Holds data being **read from or written to memory**.
- All memory data passes through the MBR.

MAR (Memory Address Register)

- Holds the **address** of the memory location to be accessed.

IR (Instruction Register)

- Holds the **8-bit opcode** of the instruction currently being executed.

IBR (Instruction Buffer Register) ⚠️ ***Very Important***

- Temporarily stores the **right-hand instruction**
- Used when the left instruction is executed first

📌 **Common Exam Question:**
IBR stores the **right instruction**, NOT the left.

PC (Program Counter)

- Holds the address of the **next instruction** to be fetched from memory.

AC (Accumulator) & MQ (Multiplier Quotient Register)

- Used for arithmetic operations in the ALU.

MQ Register:

- Stores:
    - Quotient in division
    - Least Significant Bits (LSB) in multiplication

AC Register:

- Stores:
    - Results of addition and subtraction
    - Most Significant Bits (MSB) in multiplication

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4. Instruction Cycle

The instruction cycle consists of **two main phases**:

1. Fetch Cycle

- PC → MAR
- Memory → MBR
- If instruction is right-hand → stored in IBR
- Opcode → IR

- Address → MAR

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2. Execute Cycle

- Control Unit decodes the instruction
- Sends control signals to execute:
  - Data transfer
  - Arithmetic or logic operation

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5. Key Exam Notes (Lecture 4)

- IBR always stores the **right instruction**
- Architecture = WHAT, Organization = HOW
- Multiplication and division require **both AC and MQ**
- Register functions are asked **by name**
